Jason B. Forsyth

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Education

Ph.D., Computer Engineering, May 2015
Virginia Polytechnic Institute and State University, Blacksburg, VA
Dissertation Title: *Exploring Electronic Storyboards as Interdisciplinary Design Tools for Pervasive Computing*
Advisor: Dr. Thomas L. Martin

M.S., Computer Engineering, May 2010
Virginia Polytechnic Institute and State University, Blacksburg, VA
Thesis Title: *Wearable Pulse Oximetry in Construction Environments*
Advisor: Dr. Thomas L. Martin

B.S., Computer Engineering, December 2007
Minors: Mathematics, Computer Science, Religion and Culture
Virginia Polytechnic Institute and State University, Blacksburg, VA

Academic Appointments

Associate Professor, Department of Engineering, James Madison University (Aug. 2021 – Present)

Assistant Professor, Department of Engineering, James Madison University (Aug. 2018 – August 2021)

Assistant Professor, Department of Engineering and Computer Science, York College of Pennsylvania (Aug. 2015 – Aug. 2018)

Special Appointments

Curator of Coins for the Madison Art Collection (2023 – Present)

Oliver Professor for Artificial Intelligence, Department of Engineering (2025 - 2026)

Faculty Affiliate, Office of Creative Propulsion, College of Visual and Performing Arts (2025 – 2026)

Honors and Awards

Best Paper award from *IEEE Transactions on Automation Science and Engineering* for "Feasibility of Intelligent Monitoring of Construction Workers for Carbon Monoxide Poisoning" (2012)

Best Paper nominee from *IEEE International Symposium on Wearable Computers* for "An Interdisciplinary Undergraduate Design Course for Wearable and Pervasive Computing Products" (2011)

William Blackwell Award from Virginia Tech ECE Department for "Feasibility of Intelligent Monitoring of Construction Workers for Carbon Monoxide Poisoning" (2011)

Nomination for Leading Lights New River Valley Service Award for middle school tutoring/mentoring program (2012)

Nomination for Paul E. Torgersen Graduate Student Research Excellence Award, Virginia Tech (2010)

Research and Scholarship

Research Interests and Experience

Wearable Computing: developing motion-sensing and haptic feedback systems that provide real-time guidance and assessment of human movement across physical rehabilitation, exercise, and performance applications.

Computational Numismatics: applying machine learning and computer vision to ancient coin identification, provenance recovery, and digitization of numismatic collections.

<u>Assistant/Associate Professor</u>, James Madison University (August 2018 – Present)

- Co-Director of JMU Wearable Computing Group (Wear@JMU) and COmputational methods In Numismatics (COIN) Group.
- Developing motion-sensing and haptic feedback systems that provide real-time guidance and assessment of human movement across physical rehabilitation, exercise, and performance applications. [J1, J2, P1, P2, P5, P9, P11, P14, P18].
- Applying deep metric learning and computer vision to ancient coin identification and provenance recovery in collaboration with the JMU Department of Computer Science and the Madison Art Collection.

<u>Assistant Professor</u>, York College of Pennsylvania (August 2015 – August 2018)

- Received two grants to create community-based multidisciplinary capstone project; grant also supports research on project's impact on engineering student efficacy, improvement of K-8 student science outcomes, and departmental recruitment of underrepresented populations [P16, C11, C12]

<u>Research Assistant</u>, ECE Department, Virginia Tech, (August 2009 – May 2015)

- Created prototyping exercises and design tools to improve collaboration between Computer Engineering, Industrial Design, and Marketing students for pervasive computing products. [J5, C16, C18, C19, P19].
- Developed wearable safety system for construction and roadside workers. [C15, C20, B2].

<u>Graduate Assistant</u>, Virginia Tech Institute for Creativity, Arts, and Technology, (August 2013 – May 2015)

- Creator and instructor for STEAM (Science, Technology, Engineering, Arts, and Math) educational experiences for K12 students [C17, P20].

External Funding (James Madison University)

Sol Lim (Virginia Tech), **Jason Forsyth** (**Co-PI**), and Michael Stewart (JMU), "Supporting College Students with Impaired Vision in a Shared Housing Through Haptic-Guided Peripersonal Navigation", from 4VA, $30,000 over 1 year. Awarded 10/17/2022 [P9]

Homero Murzi (Virginia Tech), **Jason Forsyth** (**Co-PI**), and Diana Franco (UVA), "Exploring Students Perceptions of Engineering Using Arts-informed Methods: A Multi-Case Study", from 4VA, $25,000 over 1 year. Awarded 11/30/2021 [C1, C6]

Jason Forsyth (PI), Tom Martin (Virginia Tech), "Designing Wearable Systems for Movement Correction in Physical Therapy Applications", from 4VA, $10,000 over 1 year. Awarded 10/31/2021 [P11]

Internal Funding (James Madison University)

Jason Forsyth, "Building Capacity for Undergraduate Research Through Conference Presentations and Faculty Training", $4,200 from College of Integrated Science and Engineering Faculty Development Fund. Awarded 12/8/2022 [P9]

Jason Forsyth, $750 from College of Integrated Science and Engineering Mini-Grant. Awarded 3/22/2022.

Jason Forsyth, Heather Kirkvold, "Designing Wearable Systems for Movement Correction in Physical Therapy Applications", $5,000 from College of Integrated Science and Engineering Faculty Development Fund. Awarded 12/17/2021 [P11]

Jason Forsyth, “Prototyping Wearables for Physical Therapy”, $690 from College of Integrated Science and Engineering Mini-Grant. Awarded 3/24/2021.

Jason Forsyth, Michael Stewart, “A Wearable Computing System to Increase Access to Healthcare and Patient Outcomes in Physical Rehabilitation Exercises”, $2,500 from College of Integrated Science and Engineering Faculty Development Fund. Proposal supports equipment purchases and student researchers to a novel wearable computing system for use in physical therapy and rehabilitation exercises. Awarded 2/1/2019. Related publications [P12, P14, P15, P18].

External Funding (York College of Pennsylvania)

Jason Forsyth, Nicole Hesson, “Developing an Automated Greenhouse at Alexander D. Goode Elementary School to Serve as a Living Laboratory for Science Education”, $9,746 over 12 months from the York County Community Foundation. Proposal supports research and capital expenses related to the automated greenhouse capstone project. Awarded 1/1/2017. Related publications: [C11, C12]

Internal Funding (York College of Pennsylvania)

Jason Forsyth, Nicole Hesson; “Addressing Economic and Educational Disadvantages in York City through Engineering Design Projects”, $17,000 over 12 months from the Office of the President. Proposal supports research investigating the impact of service-based multidisciplinary projects on engineering students, faculty, and community partners. Also, provides material support for capstone projects to develop an automated greenhouse, assistive technologies, and targeted curricula for K-8 students. Awarded: 3/24/2016. Related publications: [C11, C12]

Book Chapters

B1. **Jason Forsyth**, “Measurements and Experiments – Data and Decisions with Sensors, Radios, and Arduinos”, Chapter 2 in *Discovering Engineering Design in the 21st Century: An Activities-Based Approach*, Cengage Publishing, 2023. ISBN: 978-0-357-68250-4

B2. R. Younes, K. Hines, **J. Forsyth**, J. Dennis, T. Martin, and M. Jones, “The design of smart garments for motion capture and activity classification,” Chapter 27 in *Smart Textiles and Their Applications*, V. Koncar (editor), Woodhead Publishing, Duxford, UK, 2016, pp. 627-655

Journal Publications

J1. Mahdis Tajdari, **Jason Forsyth**, and Sol Lim, “Navigating with Haptic Gloves: Investigating Strategies for Horizontal and Vertical Movement Guidance” in International Journal of Human - Computer Studies, Volume 203, September 2025

J2. Mahdis Tajdari, **Jason Forsyth**, and Sol Lim, “Sensitivity to Vibrotactile Stimulation in the Hand and Wrist: Effects of Motion, Temporal Patterns, and Biological Sex” in Human Factors: The Journal of the Human Factors and Ergonomics Society, Volume 64, Issue 4, pp. 331-345, 2024

J3. Nathan Tenhundfeld, **Jason Forsyth**, Nathan Sprague, Samy El-Tawab, Jenna Cotter, and Lisa Vangsness, “In the rough: Evaluation of convergence across trust assessment techniques using an autonomous golf cart”, Journal of Cognitive Engineering and Decision Making, Volume 18, Issue 1, pp. 3-21, 2024

J4. **Jason Forsyth**, Tom Martin, Deborah Young-Corbett, Ed Dorsa, “Feasibility of Intelligent Monitoring of Construction Workers for Carbon Monoxide Poisoning”, *IEEE Trans. on Automation Science and Engineering*, Volume 9, Issue 3, pp. 505-515, July 2012 (16% acceptance rate) **(2012 Best Paper Award)**

J5. Tom Martin, Kahyun Kim, **Jason Forsyth**, Lisa McNair, Eloise Coupey, Ed Dorsa, “Discipline-Based Instruction to Promote Interdisciplinary Design of Wearable and Pervasive Computing Products”, *Personal and Ubiquitous Computing*, Volume 17, Issue 3, pp. 465-478, 2013

Peer-Reviewed Conference Publications (Full Papers)

C1. Homero Murzi, Diana Franco, **Jason Forsyth**, Karen Martinez Soto, Matthew James, and Lisa Schibelus, “WIP: Developing an arts-informed approach to understand students’ perceptions of engineering”**,** Proceedings of the IEEE Frontiers in Education (FIE), Uppsala, Sweden, October 8-11, 2022

C2. Michael S. Thompson, Alan Cheville, **Jason Forsyth**, “Addressing Convergent Problems with Entrepreneurially-Minded Learning”, Proceedings of the American Society of Engineering Education Annual Conference – Entrepreneurship & Engineering Innovation Division (ENT), Minneapolis, MN, June 26-29, 2022

C3. Zixuan Yan, Rahib Younes, **Jason Forsyth**, “ResNet-like CNN Architecture and Saliency Map for Human Activity Recognition”, EAI MobiCASE 2021 – November 2021

C4. Kurt Patterson, Justin Henriques, Daniel Castaneda, Robert Nagel, Kyle Gipson, Shraddha Joshi, Callie Miller, Jacquelyn Nagel, and **Jason Forsyth**, “An ecosystem to support sense-making, identity formation, and belonging for first-year engineering students”, 12th Annual First Year Engineering Experience (FYEE) Conference, Virtual, August 9-10th 2021

C5. Daniel Castaneda, **Jason Forsyth**, Shraddha Joshi, Callie Miller, Kyle Gipson, Jacquelyn Nagel, Robert Nagel, Kurt Paterson, “24for24: A Virtual Summer Bridge Program in Multiple 24-Minute Sessions for the Collegiate Class of 2024”, 2020 Sixth International Conference on e-Learning (econf), Virtual, December 2020

C6. Michael James, Homero Murzi, **Jason Forsyth**, Lilianny Virguez, Pamela Dickrell, “Exploring Perceptions of Disciplines using Arts-Informed Methods”, Proceedings of the American Society of Engineering Education Annual Conference – First Year Program Division, Virtual, June 2020

C7. Michael Stewart, **Jason Forsyth**, Zamua Nasrawt, “EduGit: Toward a Platform for Publishing and Adopting Course Content” at the 21st International Conference of Human-Computer Interaction (HCII), Orlando, FL, July 26-31, 2019.

C8. Robert Nagel, Jacquelyn Nagel, Callie Miller, **Jason Forsyth**, Shraddha Joshi, Kyle Gipson, “Engagement in Practice: Engaging with the Community One Bike at a Time”, Proceedings of the American Society of Engineering Education Annual Conference – Community Engagement Division, Tampa Bay, FL, June 16-19, 2019

C9. Kala Meah, James Moscola, Jim Kearns, Eleanor Leung, **Jason Forsyth**, “A Seven-week Module to Introduce Electrical and Computer Engineering to Freshmen Engineering Students”, Proceedings of the American Society of Engineering Education Annual Conference – First-Year Programs Division, Tampa Bay, FL, June 16-19, 2019

C10. Kala Meah, **Jason Forsyth**, James Moscola, “A Smart Sensor Network for an Automated Urban Greenhouse”, 2019 International Conference on Robotics, Electrical and Signal Progressing Techniques (ICREST), Dhaka, Bangladesh, January 10-12, 2019

C11. **Jason Forsyth**, Mark Budnik, Randi Shedlosky, Jeff Will, “Effects of Service-Learning Projects on Capstone Student Motivation”, Proceedings of the American Society of Engineering Education Annual Conference – Multidisciplinary Engineering Division, Salt Lake City, Utah, June 24-27, 2018

C12. **Jason Forsyth**, Nicole Hesson, “Benefits and Challenges Transitioning to Community Service Multidisciplinary Capstone Projects”, Proceedings of the American Society of Engineering Education Annual Conference – Multidisciplinary Engineering Division, Columbus, Ohio, June 25-28, 2017

C13. Stephen Wilkerson, **Jason Forsyth**, Christopher Korpela, “Project-Based Learning Using the Robotic Operating System (ROS) for Undergraduate Research Applications”, Proceedings of the American Society of Engineering Education Annual Conference – Multidisciplinary Engineering Division, Columbus, Ohio, June 25-28, 2017

C14. Stephen Wilkerson, **Jason Forsyth**, Cara Sperbeck, Matthew Jones, Patrick Lynn, “A Student Project Using Robotic Operating System (ROS) for Undergraduate Research”, Proceedings of the American Society of Engineering Education Annual Conference – Computing & Information Division, Columbus, Ohio, June 25-28, 2017

C15. **Jason Forsyth**, Tom Martin, Darrell Bowman, “Feasibility of GPS-based Warning System for Roadside Workers”, IEEE International Conference on Connected Vehicles and Expo, Vienna, Austria, November 3-7, 2014 (33% acceptance rate)

C16. **Jason Forsyth**, Tom Martin, “Extracting Behavioral Information from Electronic Storyboards”, Proceedings of the 6th ACM SIGCHI Symposium on Engineering Interactive Computer Systems, Rome, Italy, June 17-20, 2014 (18% acceptance rate)

C17. Blake Sawyer, **Jason Forsyth**, Taylor O’Connor, Brennon Bortz, Teri Finn, Liesl Baum, Ivica Ico Bukvic, Benjamin Knapp, Dane Webster, “Form, Function, and Performances in a Musical Instrument MAKErs Camp”, Proceedings of the 44th SIGCSE Technical Symposium on Computer Science Education, Denver, CO, March 6-9, 2013, pp. 669-674 (37.8% of 293 accepted as long papers)

C18. Lisa McNair, Kahyun Kim, **Jason Forsyth**, Ed Dorsa, Tom Martin, Eloise Coupey, “Interdisciplinary Pedagogy for Pervasive Computing Design Processes: An Evaluative Analysis,” American Society of Engineering Education Annual Conference. San Antonio, TX, June 11, 2012

C19. Tom Martin, Kahyun Kim, **Jason Forsyth**, Lisa McNair, Eloise Coupey, Ed Dorsa, “An Interdisciplinary Undergraduate Design Course for Wearable and Pervasive Computing Products”, 2011 IEEE International Symposium on Wearable Computers (ISWC), San Francisco, CA, June 12-15 2011, pp.61-68 **(Best Paper Nominee)** (19% of 31 accepted as long papers)

C20. **Jason Forsyth**, Tom Martin, Deborah Young-Corbett, Ed Dorsa; "Feasibility Study of a Wearable Carbon Monoxide Warning System for Construction Workers," 2011 IEEE International Conference on Pervasive Computing and Communications (PerCom), Seattle, WA, March 21-25 2011, pp.28-36 (11% of 156 accepted as long papers)

Peer-Reviewed Conference Posters, Panels, and Presentations
** indicates student author*

P1. Julia Larson*, **Jason Forsyth**, “Do You Feel the Burn? Exploring Lightweight Haptic Feedback and How It Can Improve Exercise Performance”, 2026 National Conference on Undergraduate Research (NCUR), April 13–15, 2026 (refereed as abstract)
P2. Julia Larson*, **Jason Forsyth**, “Do You Feel the Burn? Exploring Lightweight Haptic Feedback and How It Can Improve Exercise Performance”, 2026 ACM Capital Region Celebration of Women in Computing (CAPWIC), March 27–28, 2026 (refereed as abstract)
P3. Dhanshree Atre*, Nathan Sprague, **Jason Forsyth**, and Trevor Schonbrun*, “Deep Metric Learning for Ancient Coin Identification”, 2026 ACM Capital Region Celebration of Women in Computing (CAPWIC), March 27–28, 2026 (refereed as abstract)
P4. Trevor Schonbrun*, Nathan Sprague, and **Jason Forsyth**, “Deep Metric Learning for Ancient Coin Identification”, Virginia Academy of Science, Engineering, and Medicine AI Summit, Roanoke, VA, September 30–October 1, 2025 (refereed as abstract)
P5. Julia Larson*, **Jason Forsyth**, initial study design presentation, Virginia Academy of Science Annual Meeting, Charlottesville, VA, May 22, 2025 (refereed as abstract)
P6. Megan Caulfield*, **Jason Forsyth**, Laura Deportes, and Daniel Castaneda, “Braille Learning using Haptic Feedback”, 2024 Systems and Information Engineering Design Symposium (SIEDS), Charlottesville, VA, USA, 2024 (published as full paper, reviewed as abstract)

P7. Sanda Thura*, **Jason Forsyth**, Kevin Molloy, and Jacob Couch, “Analysis of the Machine Learning Classification of Cardiac Disease on Embedded Systems” 2023 Systems and Information Engineering Design Symposium (SIEDS), Charlottesville, VA, USA, 2023 (published as full paper, reviewed as abstract)

P8. Ritvash Chowdhury*, Michael Chung*, Dylan Gnagey*, and **Jason Forsyth**, “GymSense: Exploration in Measuring Bench Press Form Using Sensor Technology” 2023 Systems and Information Engineering Design

Symposium (SIEDS), Charlottesville, VA, USA, 2023 (published as full paper, reviewed as abstract)

P9. Justin Blevins*, Megan Caulfield*, and **Jason Forsyth**, “Tactile Instructions for Wearable Physical Rehabilitation”, Poster Presentation at the 2023 National Conference on Undergraduate Research (NCUR), Eau-Claire, WI, USA 2023

P10. Jenna E. Cotter, Emily H. O’Hear, R. Cooper Smitherman, Addison B. Bright, Nathan L. Tenhundfeld, **Jason Forsyth**, Nathan R. Sprague, Samy El-Tawab, "Convergence Across Behavioral and Self-Report Measures Evaluating Individuals Trust in an Autonomous Golf Cart", 2022 Systems and Information Engineering Design Symposium (SIEDS), Charlottesville, VA, USA, 2022 (published as full paper, reviewed as abstract)

P11. Megan Caulfield*, **Jason Forsyth** “Student Research Short: Wearable Computing for Physical Rehabilitation”, Presentation at 2022 ACM Capital Region Celebration of Women in Computing (CAPWIC) (refereed as abstract)

P12. Stephen Mitchell*, **Jason Forsyth**, Michael S. Thompson, “Exploring Amateur Performance in Athletic Tests Using Wearable Sensors” 2021 Systems and Information Engineering Design Symposium (SIEDS), Charlottesville, VA, USA, 2021 (published as full paper, reviewed as abstract)

P13. Jorge Barajas, Christian Detweiler, Cailyn Lager, Charles Seaver, Mark Vakrchuk, Justin Henriques, **Jason Forsyth**, “A Toolkit for the Spatiotemporal Analysis of Eutrophication Using Multispectral Imagery Collected from Drones” 2021 Systems and Information Engineering Design Symposium (SIEDS), Charlottesville, VA, USA, 2021 (published as full paper, reviewed as abstract)

P14. Sophia Cronin*, Tyler Webster*, **Jason Forsyth** “Student Research Short: Wearable Computing for Physical Rehabilitation”, Presentation at 2021 ACM Capital Region Celebration of Women in Computing (CAPWIC) (refereed as abstract)

P15. Stephen Mitchell*, **Jason Forsyth**, Michael Thompson, “Assessing Athletic Performance with a Wearable Inertial Measurement Unit”, Presentation at 2021 IEEE South East Conference, Virtual. (refereed as a short paper)

P16. Nicole Hesson , **Jason Forsyth**, “Effects of Informal versus School-Based Field Experience on Elementary Pre-service Teachers' Self-Efficacy for Teaching Science”, Presentation at the 2020 National Association for Research in Science Teaching (NARST) International Conference, Portland, Oregon (refereed as short paper)

P17. I. J. Miller, B. Schieber, Z. De Bay, E. Benner, J. Ortiz, J. Girdner, P. Patel, D. Coradazzi, J. Henriques, **J. Forsyth**, "Analyzing crop health in vineyards through a multispectral imaging and drone system," 2020 Systems and Information Engineering Design Symposium (SIEDS), Charlottesville, VA, USA, 2020 (published as full paper, reviewed as abstract)

P18. Sanarea Ali*, **Jason Forsyth** “Student Research Short: Wearable Computing for Assessing Joint Angles and Range of Motion”, Presentation at 2019 ACM Capital Region Celebration of Women in Computing (CAPWIC) (refereed as abstract)

P19. **Jason Forsyth**, “Using Electronic Storyboards to Support Interdisciplinary Design of Pervasive Computing Systems”, Poster Presentation for the Sixth Annual Ph.D. Forum on Pervasive Computing and Communications (PerCom), San Diego, CA, March 20, 2013

P20. Jamie Simmons, Liesl Baum, **Jason Forsyth**, “Engaging Middle School Students in SEAD Learning Through Smart Product Design”, Conference on Creativity and Fabrication in Education, Oct 25-26, 2014, Stanford, CA (refereed short paper for panel discussion)

Other Publications

O1. Ed Dorsa, Eloise Coupey, Tom Martin, Lisa McNair, **Jason Forsyth**, Sophie Kim, "Design Thinking Meets Computational Thinking – An Interdisciplinary Exercise in Developing Smart Product", Industrial Design Society of America Education Symposium, August 2012

O2. **Jason Forsyth**, Tom Martin, "Tools for Interdisciplinary Design of Pervasive Computing", *International Journal of Pervasive Computing and Communications*, Volume 8, Issue 2, pp. 112-132, June 2012 (Invited Survey)

O3. Tom Martin, Eloise Coupey, Lisa McNair, Ed Dorsa, **Jason Forsyth**, Sophie Kim, Ron Kemnitzer, "An Interdisciplinary Design Course for Pervasive Computing", *IEEE Pervasive Computing*, Volume 11, Issue 1, pp.80-83, January 2012

Invited Talks and Panels

1. J. Forsyth, "Career Landscape Workshop Series: What is the landscape of teaching-oriented careers?". Sponsored by CRA-E. May 16th, 2024

2. J. Forsyth, "Teaching Oriented Academic Careers: The Landscape is Broader than You Think", ACM Federated Computing Research Conference (FCRC). Sponsored by CRA-E. Orlando, FL. June 21st, 2023

3. J. Forsyth, "Computing as a Design and Artistic Material", Research Seminar for the JMU Office of Creative Propulsion, College of Visual and Performing Arts, October 21st, 2022

4. J. Forsyth, "Applications of Wearable Computing for Physical Therapy and Rehabilitation", Research Seminar for the JMU Computer Science Department, February 22nd, 2019

Creative Works and Exhibitions

1. Take 499 – An interactive architecture installation at the York College Faculty Biennale in Fall 2016. A collaboration with Matthew Campbell (York College Department of Arts and Communication) explores various notions of "time" through music, mechanics, and the prime number distribution (video).

Graduate Research Students / Advising

PhD Committee Member – Mahdis Tajdari, PhD in Industrial Systems Engineering, Virginia Tech. Graduated December 2025.

Undergraduate Independent Study / Research Students

Current Students

Sammy Nelms (ENGR), Topic: Hardware Design to Support Novel Haptic Feedback (Spring 2024 – present)

Trevor Schonbrun (ENGR), Topic: Deep Metric Learning for Ancient Coin Identification. (Fall 2025 – present)

- Trevor Schonbrun*, Nathan Sprague, and **Jason Forsyth**, "Deep Metric Learning for Ancient Coin Identification", Virginia Academy of Science, Engineering, and Medicine AI Summit, Roanoke, VA, September 30–October 1, 2025 (refereed as abstract)

Graduated Students

Julia Larson (ENGR), Honors Thesis: Haptic Feedback Inspired by Physical Therapy Practice for Exercise Performance. (Fall 2023 – Spring 2026)

- Julia Larson*, **Jason Forsyth**, "Do You Feel the Burn? Exploring Lightweight Haptic Feedback and How It Can Improve Exercise Performance", 2026 National Conference on Undergraduate Research (NCUR), April 13–15, 2026 (refereed as abstract)
- Julia Larson*, **Jason Forsyth**, "Do You Feel the Burn? Exploring Lightweight Haptic Feedback and How It Can Improve Exercise Performance", 2026 ACM Capital Region Celebration of Women in Computing (CAPWIC), March 27–28, 2026 (refereed as abstract)

- Julia Larson*, **Jason Forsyth**, "Do You Feel the Burn? Exploring Lightweight Haptic Feedback and How It Can Improve Exercise Performance", Virginia Academy of Science Annual Meeting, Charlottesville, VA, May 22, 2025 (refereed as abstract)

Dhanshree Atre (CS), Topic: Deep Metric Learning for Ancient Coin Identification. (Fall 2025 – Spring 2026)
- Dhanshree Atre*, Nathan Sprague, **Jason Forsyth**, and Trevor Schonbrun*, "Deep Metric Learning for Ancient Coin Identification", 2026 ACM Capital Region Celebration of Women in Computing (CAPWIC), March 27–28, 2026 (refereed as abstract)

Qunnie Li, First-Year Research Student: Development of Image and Video Dataset for Ancient Coin Identification. (Fall 2025 – Spring 2026)

Jackson Greer (CS), Topic: Deep Metric Learning for Ancient Coin Identification. (Spring 2024 – Spring 2025)
- Presented at JMU Computer Science Research Day, Harrisonburg, VA, April 25, 2025

Maria Lo Presti, HON 450: 3D Modeling and Fabrication for Digital Numismatics. (Fall 2024)
- Presented numismatic research on Sicilian coinage at JMU Honors Symposium, Fall 2024

Megan Caulfield, Honors Thesis: *Braille Learning using Haptic Feedback* Readers: Dr. Laura Deportes (Education), Dr. Daniel Castaneda (Engineering). (Fall 2021 – Spring 2024)
- Megan Caulfield*, **Jason Forsyth**, Laura Deportes, and Daniel Castaneda, "Braille Learning using Haptic Feedback", 2024 Systems and Information Engineering Design Symposium (SIEDS), Charlottesville, VA, USA, 2024 (published as full paper, reviewed as abstract) [P6]

Will Bradford, Topic: Design and Revision of a Wearable Haptic Drive System (Fall 2022 – Spring 2024)

Sanda Thura (CS), Honors Thesis: *Analysis of the Machine Learning Classification of Cardiac Disease on Embedded Systems* Readers: Dr. Kevin Molloy (CS), Dr. Jacob Couch (Johns Hopkins) (Spring 2022 – Spring 2023)
- Sanda Thura*, **Jason Forsyth**, Kevin Molloy, and Jacob Couch, "Analysis of the Machine Learning Classification of Cardiac Disease on Embedded Systems" 2023 Systems and Information Engineering Design Symposium (SIEDS), Charlottesville, VA, USA, 2023 (published as full paper, reviewed as abstract) [P7]

Stephen Mitchell, Topic: Automatic Detection and Assessment of Athlete Performance during a Drop Jump Test. *(Research in collaboration with Bucknell University)* (Fall 2018 – Spring 2022).
- Presented at the 2019 Mid-Atlantic Regional Conference on Undergraduate Scholarship (MARCUS) at Randolph College
- Poster presentation at the 2021 IEEE South East Conference (refereed as short paper) [P15]
- Full paper and presentation at SIEDS 2021 [P12]

Maxwell Stevens, Topic: Prediction of Human Motion Using Robotic Planning Algorithms (Fall 2022 – Spring 2023)

Riley White, Topic: Automatic Decomposition of Human Motion and Activities (Fall 2021 – Spring 2023)

Dawson Dolansky, Topic: Integrating Data Collection from Inertial Measurement and Video Capture (Fall 2021 – Spring 2023)

Tyler Webster, Topic: Exploration of visual and light feedback for activity correction and reinforcement in physical therapy exercises. (Fall 2019 – Spring 2022)
- Presented at the 2021 ACM Capital Area Conference on Women in Computing (CAPWIC) [P14]

Jonathan Li, Topic: Development of 3D modeling forms to render physical motion with inertial measurement units. (Spring 2020 – Spring 2021)

Sophia Cronin, Topic: Exploration of haptic feedback for activity correction and reinforcement in physical therapy exercises. (Fall 2019 – Spring 2021)

- Presented at the 2021 ACM Capital Area Conference on Women in Computing (CAPWIC) [P14]

Sanarea Ali, Topic: Using wearable inertial measurement units to support visualization and feedback mechanisms for physicians supervising patients undergoing physical therapy applications. (Graduated 2020)

- Presented at the 2019 ACM Capital Area Conference on Women in Computing (CAPWIC)

Annie (Ernie) Benner, Topic: Development of Software Tools to Support Activity Recognition in Physical Therapy Applications (Graduated 2020)

<u>Summer 2018</u>
Amanda Redhouse, Topic: Application of Convolution Neural Networks for On-body Activity Recognition

<u>Spring 2018</u>
Ashleen Hayes and Amanda Redhouse. Topic: Development of on-body sensing and human activity recognition for physical therapy applications.

<u>Summer 2017</u>
Shawn O'Brien. Topic: Exploration of wireless sensor network for on-body wearable activity recognition.

<u>Spring 2017</u>
Nicholas Whorley, Benjamin Newlin. Topics: Development of autonomous robotic systems using the Robot Operating System (ROS). Student projects included control of a robotic arm and voice control of a Turtlebot robot.

<u>Fall 2016</u>
Matthew Jones, Cara Sperbeck, Patrick Lynn. Topics: Development of autonomous robotic systems using the Robot Operating System (ROS). Student projects included: target tracking using fiducial markers and object color, and navigation using existing architectural drawings. Description of student work and experiences in [C13,C14].

Professional Activities

Conference Chairing and Organization
General Chair, IEEE Systems and Information Engineering Design Symposium (SIEDS) (2023, 2024)

Program Chair, ACM Capital Region Celebration of Women in Computing Conference (2024)

Program Committee Member, ACM Capital Region Celebration of Women in Computing Conference (2019 – present)

Virtual Platform Organizer, ACM Capital Region Celebration of Women in Computing Conference (2021, 2022)

Publications Chair, IEEE Systems and Information Engineering Design Symposium (SIEDS) (2021, 2022)

Division Chair, ASEE Engineering Innovation and Entrepreneurship (ENT) Division (2021-2022)

Program Chair, ASEE Engineering Innovation and Entrepreneurship (ENT) Division (2019-2020)

Reviewing in Wearable and Ubiquitous Computing
Reviewer, IEEE International Symposium on Wearable Computers (ISWC) (2013, 2015, 2016, 2021, 2022)
Reviewer, ACM Conference on Human Factors in Computing Systems (CHI) (2020, 2022, 2023)
Reviewer, Demo and Poster Session for ACM International Conference on Ubiquitous Computing (Ubicomp) and the International Symposium on Wearable Computers (ISWC) (2021, 2022, 2023, 2025, 2026)
Reviewer, ACM/IEEE International Conference on Ubiquitous Computing (Ubicomp) (2015, 2016)

Reviewer, ACM Conference on Designing Interactive Systems (DIS) Works-in-Progress (2014)
Student Volunteer, ACM Conference on Designing Interactive Systems, Vancouver, CA, June 2014
Student Volunteer, IEEE ISWC/Pervasive 2011, San Francisco, June 2011
Student Volunteer, IEEE PerCom 2011, Seattle WA, March 2011

Reviewing in Engineering Education
Reviewer, Journal of Engineering Education, (2023, 2024)
Reviewer, IEEE Transactions on Education (2016-2017)

Reviewer, ASEE National Conference Multidisciplinary Engineering Division (2016 - present)
Reviewer, ASEE National Conference Electrical and Computer Engineering Division (2016 - present)
Reviewer, ASEE National Conference Entrepreneurship and Engineering Innovation Division (2016 – present)
Reviewer, ASEE National Conference K-12 Division (2016)

Affiliations

Member: IEEE, HKN, ACM, ASEE

Teaching Experience

Teaching Experience (James Madison University)

Summary of Courses Taught at James Madison University

Course Title	Semester Offered
ENGR 112: Engineering Decisions	Spring 2019 – Spring 2022
ENGR 280: Real-World Modeling and Programming	Spring 2020
ENGR 265E/280/298: Python Programming with Biometric Data	Fall 2020, Spring 2022, Spring 2023
ENGR 231: Engineering Design I	Fall 2018, Fall 2020 – Fall 2025
ENGR 232: Engineering Design II	Spring 2021- present
ENGR 313: Circuits and Instrumentation	Fall 2018, Spring 2019, Fall 2019, Spring 2020, Fall 2020, Fall 2021
ENGR 315/340: Computational Methods in Engineering	Fall 2024,
ENGR 345: Fundamentals of Computer Engineering	Fall 2023, Fall 2024, Fall 2025
ENGR 445: Embedded System Design	Fall 2019, Fall 2021, Fall 2022, Spring 2024, Fall 2024, Spring 2026
ENGR 480 / CS 497: Autonomous Vehicles (co-taught with Dr. Nathan Sprague)	Spring 2021
ENG 490 – Dance and Technology (co-taught with Prof. Tara Lee Burns)	Spring 2026

Description of Selected Courses

ENGR 313 – Circuits and Instrumentation: Junior/senior course covering broad topics in DC/AC circuits and power analysis. Course prepares students for capstone work involving electronics and for the FE exam. Topics covered include: resistive circuit analysis, transistors, operational amplifiers, first-order circuits, second-order circuits, analog and digital filters, and phasor space analysis. Lecture topics are reinforced via weekly lab sessions and include two design-build projects throughout the semester.

ENGR 315 – Computational Methods in Engineering: An introduction to modern programming languages and their application within the larger engineering field. The course will focus on modeling, mathematical and statistical analysis, data visualization, and programming techniques that can be used in engineering design and decision making. Integrated code development tools will be utilized to maintain programming environments and allow for code management and collaboration.

ENGR 445 – Embedded System Design: This senior project-based course will prepare students to design novel computing systems that involve sensing, actuation and networking. Various topics will be covered including component selection, microprocessor peripheral interfaces, energy efficiency and power-aware design, and wireless communication methods. Students will design, test and fabricate their own embedded computing system to collect and analyze biometric data.

ENGR 490 – Dance and Technology: This collaborative, project-based course invites students to explore how movement and machines co-create. Working with motion capture, sensors, and interactive systems, students will create kinetic sculptures, responsive wearables, and machine-learning “partners” that move, react, and improvise alongside the human body. Through studio practice and performance experiments, students investigate embodiment, interactivity, and co-creation with technology. By the end of the semester, students will produce original works that highlight how choreography and engineering generate new forms together.

Teaching Experience (York College of Pennsylvania)

Summary of Courses Taught at York College of Pennsylvania

Course Title	Semester Offered
FYS 100 – Creative Computing	Fall 2016, 2017 (co-taught with Dr. David Hovemeyer)
EGR 100 – Engineering Practice and Design Studio	Fall 2015, 2016
ECE 220 – Design and Analysis of Digital Circuits	Fall 2015, 2016, 2017
ECE 260 – Fundamentals of Computer Engineering	Spring 2016
ECE 335 – Discrete Math with Applications	Summer 2016, 2017
ECE 370 – Microprocessor System Design	Summer 2016, 2017, 2018
ECE 400 – Capstone Design I	Summer 2016, 2017, 2018
ECE 420 – Embedded System Design	Summer 2016, 2017, 2018

Description of Selected Courses

FYS 100 – Creative Computing: First year seminar engages freshman students in employing computing systems to develop artistic and creative applications. Students utilize Processing, Arduino, and Raspberry Pi computers to create interactive installations as a final project. Hands-on labs, contemporary readings, and class discussions center around the broader them of “Can computers create art?”

EGR 100 – Engineering Practice and Design Studio: Introductory engineering course for freshman students shared between the Mechanical and Electrical and Computer Engineering programs. Students engage in hands-on learning with basic electronics and computing systems to develop design and debugging skills. Final project uses a small robotic platform and an Arduino to create a line-following robot.

ECE 220 – Design and Analysis of Digital Circuits: Sophomore course introducing electrical and computer engineering students to digital circuit design including combinational and sequential circuits. Associated lab section provides instruction in circuit design using discrete ICs as well as projects in Verilog HDL.

ECE 370 – Microprocessor System Design: Junior combined lecture/lab course providing in-depth instruction on microprocessor architecture, organization, and associated peripherals. Students utilize Silicon Labs ARM processor and C++ to design energy efficient applications using a variety of external sensors. Topics on PCB design, embedded tool chains, and processor initialization are also covered.

ECE 400 – Capstone Design I: Multidisciplinary design experience for electrical, computer, and mechanical engineers. Projects provide a substantial design, build, and test experience over a one year period. Recent projects have focused on community service and include development of an automated greenhouse for a local school, prototype medical devices, and assistive technologies for persons with mental and physical disabilities.

ECE 420 – Embedded System Design: Culminating course for electrical and computer engineering students combining all aspects of microprocessor systems. Semester long project focuses on development and optimization of wearable pedometer system. Each student must conduct research towards a selected optimization goal of energy efficiency, step count accuracy, or physical size minimization.

Teaching Experience (Virginia Tech)

Teaching Assistant, CS 2984/Arch 3514: Introduction to Physical Computing, Virginia Tech, (Spring 2014)

- Prepared electronics prototyping exercises and delivered lectures on user experience research for interdisciplinary computer science and architecture course.
- Mentored students one-on-one and in teams. Final projects were internally and externally sponsored from NASA, Virginia Tech Transportation Institute, and Virginia Tech Institute for Creativity, Arts, and Technology.

Outreach Instructor, North Cross Middle School, Roanoke, VA (September 2013 – December 2013)

- Developed 13-week product design course for middle school students through the Virginia Tech Institute for Creativity, Arts, and Technology.
- Instruction in market research, sketching, storyboarding, electronic and physical prototyping, and business plan development.
- Final presentations to local business leaders and venture capitalists.

Teaching Assistant, ECE 4974: Interdisciplinary Product Design, Virginia Tech (Fall 2009, 2010, 2011, 2012)
- Developed prototyping exercise for marketing, industrial design, and computer engineering students.
- Advised students on design and implementation of final projects.
- Experience formed the basis for dissertation research on interdisciplinary prototyping of pervasive systems.

Coordinator, Creating Interactive Prototypes Workshop, IDSA Southern Conference, Atlanta GA, (April 2012)
- Taught a half day workshop on interactive prototyping to students and practitioners at a regional industrial design conference.

Instructor, Interactive Architecture Course and Exhibition, Virginia Tech, (Spring 2011)
- Taught five workshops on interactive architecture for 3rd year architecture and interior design students.
- Workshops met once every two weeks to design new interactions based around Arduino microcontrollers and embedded sensors. Each workshop focused on a different physical "sense" such as light, sound, and touch.
- Final projects were developed over one week and exhibited in the department for 15% of semester studio grade.

Director, Fun143 Tutoring Program, Blacksburg VA (August 2008 – December 2011)
- Program sought to increase educational and personal outcomes for disadvantaged middle/high school students.
- Addressed social and economic needs of students and families by providing weekly meals, an open community, and assistance with school supplies, clothes, and other household needs.

College, Departmental, and Community Service and Outreach

National Service

Division Chair for ASEE Engineering Innovation and Entrepreneurship (ENT) Division (2021-2022)
Division Chair-Elect for ASEE Engineering Innovation and Entrepreneurship (ENT) Division (2020-2021)
Program Chair for ASEE Engineering Innovation and Entrepreneurship (ENT) Division (2019-2020)
Program Chair-Elect for ASEE Engineering Innovation and Entrepreneurship (ENT) Division (2018-2019)

College Service

Member, CISE Faculty Council (Fall 2023 – present): CISE-level advisory board representing the interests of CISE faculty and facilitating frequent communication with the CISE Dean on college-wide matters.
Member, JMU Libraries Faculty Advisory Board (2024 – Present): Advisory board that meets several times annually to advise the Dean of Libraries on policy matters and relay faculty concerns.

Member, CISE Faculty Leaves and Grants (FLAG) Committee (Fall 2022 – Spring 2023): CISE-level committee examines faculty professional development grants, leaves, and awards.

Member, York College Faculty Development Funding Committee (Fall 2016 – August 2018): Committee of the academic senate funding faculty research, travel, and leadership development grants.

Department Service

Search Chair, Teaching Faculty Positions (Visiting Lecturer and Post-Doctoral Teaching) (2025–2026)

Engineering Honors Liaison with JMU Honors College (August 2021 – Present)

Advisor, Virginia Tech – James Madison University Tech Talent Pipeline (June 2020 – Present)

Member, Department of Engineering Curriculum and Instruction Committee (C&I) (2021 – 2024)

Member, Department of Engineering Personnel Advisory Committee (PAC) (2019 – 2020, 2023 – 2026)

First Century and Madison Scholar Scholarship Reviewer (Spring 2024 – Present)

Member, Tenure Track Faculty Search Committee (Fall 2022 – Spring 2023)

Member, Engineering First Year Experience Committee (Fall 2017): Departmental committee focusing on improvement of first year engineering student experiences.

Member, Engineering Expansion Committee (Spring 2017): Departmental committee examining possible new majors and programs for expansion.

Electrical Engineering Faculty Search Committee (Fall 2015 – Spring 2016; Fall 2017 – Spring 2018)

Community Service and Outreach

JMU STEM Center Outreach (2021 – 2025): Conducted several workshops with sensors and Arduinos on environmental data collection and analysis for high school students.

Tech Tinker K-12 Outreach (August 2016 – June 2018): Monthly meet-up for K-12 students and parents to explore STEM topics and engage with YCP faculty and students.

Tech Girlz (April 9th, 2016): Full day workshop at Martin Library (York, PA) in collaboration with United Fiber and Data. Provided workshops for young girls on “tech” topics such as: physical computing with Arduino, circuits with LittleBits, and photo editing with Photoshop.